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 MAT067 University of California, Davis Winter 2010

Midterm Exam
 (Wednesday, February 10)

This exam has 7 numbered pages. Notes, books and calculators are not to be used. Solutions must be clearly written otherwise some points may be deducted even if you get the right answer. Partial credit is given only when the solutions is written clearly.

Problem	Max.	Score
1	15	
2	15	
3	15	
Total	45	

Problem 1.

a) (4 points) Suppose we are given a complex number $z = x + yi$ where $x, y \in \mathbb{R}$. Write the definitions of the complex conjugate $\bar{z}$ and the modulus $|z|$.

$$\bar{z} = x - iy$$

$$|z| = \sqrt{x^2 + y^2}$$

b) (3 points) Only based on a), prove for nonzero z that

$$w = \frac{\bar{z}}{|z|^2}$$

is an inverse of z .

$$\begin{aligned} z w &= \frac{z \bar{z}}{|z|^2} = \frac{(x+iy)(x-iy)}{x^2+y^2} \\ &= \frac{x^2+y^2}{x^2+y^2} = 1 \end{aligned}$$

$$\text{so } w = z^{-1}$$

c) (4 points) Prove that

$$e^{i\theta_1} e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$$

only by using the definition of $e^{i\theta}$ the polar form for complex numbers.

$$\begin{aligned} e^{i\theta_1} e^{i\theta_2} &= (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= \cos \theta_1 \cos \theta_2 + i \cos \theta_1 \sin \theta_2 + i \sin \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 \\ &= \cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2) \end{aligned}$$

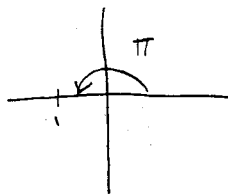
d) (4 points) Find the solutions over $\mathbb{C}$ for the equation:

$$z^3 = -1$$

$$\begin{aligned} z^3 &= -1 = e^{\pi i} \\ &= e^{(\pi + 2\pi k)i}, \quad k \in \mathbb{Z} \end{aligned}$$

$$z = e^{\frac{1}{3}(\pi + 2\pi k)i}, \quad k = 0, 1, 2$$

$$= \{e^{\frac{\pi}{3}i}, e^{\pi i}, e^{\frac{5\pi}{3}i}\}$$



Problem 2. Let V be a vector space over $\mathbb{F}$.

a) (3 points) Let U be a subset of V . State the three conditions for U to be a subspace of V .

1. $0 \in U$ U contains zero element
2. $x, y \in U \Rightarrow x + y \in U$ closed under addition
3. $x \in U, a \in \mathbb{F} \Rightarrow ax \in U$ closed under scalar multiplication

b) (2 points) Let U_1 and U_2 be subspaces of V . Which of the above conditions prevents $U_1 \cup U_2 = \{v \in V \mid v \in U_1 \text{ or } v \in U_2\}$ to be a subspace of V ?

Condition 2: $U_1 \cup U_2$ not necessarily closed under addition

$$\begin{aligned} U_1 &= \text{span}\{(0, 1)\} \\ U_2 &= \text{span}\{(1, 0)\} \end{aligned} \quad \left. \vphantom{\begin{aligned} U_1 &= \text{span}\{(0, 1)\} \\ U_2 &= \text{span}\{(1, 0)\} \end{aligned}} \right\} \Rightarrow U_1 \cup U_2 = \{(0, x), (x, 0) \mid x \in \mathbb{F}\}$$

$(1, 0), (0, 1) \in U_1 \cup U_2$ but $(1, 1) \notin U_1 \cup U_2$

c) (4 points) Let U_1 and U_2 be subspaces of V . Show that if $V = U_1 \oplus U_2$ (V is the direct sum of U_1 and U_2) then

$$U_1 \cap U_2 = \{0\}.$$

(1) $\left\{ \begin{array}{l} \text{Since } V = U_1 \oplus U_2, \quad 0 \in V \text{ is expressed uniquely} \\ \text{as } 0 + 0 = 0 \text{ (and no other way), with } 0 \in U_1 \text{ and} \\ 0 \in U_2. \end{array} \right.$

Let $u \in U_1 \cap U_2$. Then $u \in U_1$, and $u \in U_2$, and
Since U_2 is a subspace, $-u \in U_2$. so

$$u + (-u) = 0. \text{ but from (1)}$$

it follows that $u = -u = 0$.

$$\text{so } U_1 \cap U_2 = \{0\}. \quad 4$$

d) (3 points) Let $V = \mathbb{C}^4$. Define

$$U_1 = \{(a, b, c, d) \in V : d = 0\}$$

Show that U_1 is a subspace of V .

U_1 satisfies ^{all} properties from part a):

$$x, y \in U_1 \Rightarrow x = (x_1, x_2, x_3, 0), y = (y_1, y_2, y_3, 0)$$

$$a \in \mathbb{C} \quad (2) \text{ for some } x_1, x_2, x_3, y_1, y_2, y_3 \in \mathbb{C}$$

$$x + y = (x_1 + y_1, x_2 + y_2, x_3 + y_3, 0) \in U_1 \quad \checkmark$$

$$(3) ax = (ax_1, ax_2, ax_3, 0) \in U_1 \quad \checkmark$$

$$(1) (0, 0, 0, 0) \in U_1 \quad \checkmark$$

e) (3 points) Show that V is not the direct sum of U_1 and U_2 where

$$U_2 = \{(a, b, c, d) \in V : a + 2b + 3c + 4d = 0\}$$

$$\text{Let } d = 0, b = c = 1, a = -5. \quad a + 2b + 3c + 4d = 0$$

$$\text{Then } (a, b, c, d) = (-5, 1, 1, 0) \in U_2$$

and $d = 0$ so

$$(a, b, c, d) \in U_1$$

$$(-5, 1, 1, 0) \in U_1 \cap U_2 \text{ and is clearly not } 0.$$

$$\text{Then } U_1 \cap U_2 \neq \{0\} \text{ so}$$

$$V \neq U_1 \oplus U_2$$

(The direct sum does not exist in this case)

Problem 3. Suppose we have vector spaces V and W over F and a linear map $T : V \rightarrow W$.

a) (4 points) Write the definitions of $\text{null}(T)$ and $\text{range}(T)$.

$$\text{null}(T) = \{x \in V \mid T(x) = 0 \in W\}$$

$$\text{range}(T) = \{y \in W \mid T(x) = y \text{ for some } x \in V\}$$

b) (3 points) Suppose we have found a subspace U such that

$$V = \text{null}(T) \oplus U$$

$$\text{Show that } \text{range}(T) = \{T(u) : u \in U\} = X$$

clearly $X \subseteq \text{range}(T)$.
 $\nwarrow$ call this set X

$$\text{Let } y \in \text{range}(T). \text{ Then } y = T(x) \text{ for some } x \in V.$$

$$\text{Then } x = n + u \text{ for some } n \in \text{null}(T), u \in U.$$

$$T(x) = T(n + u) = T(n) + T(u) = T(u)$$

$$\text{Then } y = T(u). \text{ Thus } \text{range}(T) \subseteq X$$

c) (5 points) In the assumption of b), show that $\dim(\text{range}(T)) = \dim U$.

Let $n = \dim U$, and let $\{u_1, u_2, \dots, u_n\}$ be a basis for U .

Then by part (b), any $w \in \text{range } T$ can be expressed as

$$w = Tu = T(a_1 u_1 + a_2 u_2 + \dots + a_n u_n) = a_1 T u_1 + a_2 T u_2 + \dots + a_n T u_n$$

for some $u = a_1 u_1 + a_2 u_2 + \dots + a_n u_n \in U$.

$$\text{so } \text{range}(T) = \text{span}\{T u_j\}.$$

Suppose $w = Tu = 0$. Then $u \in \text{null}(T) \cap U = \{0\}$.

$$\text{so } u = a_1 u_1 + a_2 u_2 + \dots + a_n u_n = 0.$$

Then $a_1 = a_2 = \dots = a_n = 0$, because $\{u_j\}$ is a linearly independent set. Then $\{T u_j\}_{j=1}^n$ is linearly independent and is thus a basis for $\text{range}(T)$.

d) (3 points) By using the formula derived from b) and c) or some other methods find $\dim(\text{null}(T))$ for the map $T: \mathbb{R}^4 \rightarrow \mathbb{R}^3$.

$$T: \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \mapsto \begin{pmatrix} 1 & 1 & 4 & 3 \\ 2 & 1 & 3 & 2 \\ 3 & 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = a v_1 + b v_2 + c v_3 + d v_4$$

$$\text{Note } v_3 = 5v_2 - v_1 \quad \text{and} \quad v_4 = 4v_2 - v_1$$

$$\text{so } \text{range}(T) = \text{span}\{v_1, v_2\}$$

$$\dim(\text{range}(T)) = 2. \quad \text{Then by dimension formula,}$$

$$\boxed{\dim(\text{null}(T)) = 2.}$$